0338. Counting Bits

Easy | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/cou

Problem

- For every i in 0..n, return the number of 1 bits in i, in O(n).

Intuition

- I >> 1 has the same bits as i except the last one, so bits[i] = bits[i >> 1] + (i & 1).

Brute Force

- Popcount each i separately: O(n log n).

Optimal Approach

- Bits[0] = 0.
- For i = 1..n: bits[i] = bits[i >> 1] + (i & 1).

Worked Example

```
n = 5.

i:    0 1 2 3 4 5
bits: 0 1 1 2 1 2
bits[5] = bits[2] + 1 = 2
```

Java Solution

```java
public class CountingBits {
    public int[] countBits(int n) {
        int[] bits = new int[n + 1];
        for (int i = 1; i <= n; i++) bits[i] = bits[i >> 1] + (i & 1);
        return bits;
    }
}
```

Complexity

- Time: O(n). Space: O(n) for the output.

Pitfalls

- An alternative recurrence is bits[i] = bits[i & (i-1)] + 1.